Raja Muhammad Bilal Arshad

FAST National University (NUCES), Pakistan
0321-8633947 — bilalarshad.code@gmail.com — LinkedIn — GitHub

SUMMARY

Cybersecurity Analyst with hands-on Purple Team experience across offensive testing and defensive monitoring. Proven ability to operate SIEM platforms, investigate incidents, and deploy detection mechanisms across enterprise environments. Strong expertise in MITRE ATT&CK mapping, IDS/IPS tuning, and SOC operations, with a focus on improving detection accuracy and reducing response time.

SKILLS

Defensive Security: SIEM (Wazuh, Splunk), IDS/IPS (Snort, Suricata), SOC Operations (L1/L2), Threat Detection, Incident Response, Firewall Engineering (pfSense, OPNsense), Log Analysis, Network Security Monitoring
Offensive Security: Network Penetration Testing, Web Security (OWASP Top 10), Privilege Escalation, Active Directory Exploitation, Wi-Fi Auditing, BloodHound, Reverse Engineering (Ghidra, x64dbg), MITRE ATT&CK
Tools & Technologies: Python, Bash, Nmap, Wireshark, Tcpdump, Zeek, Sysmon, Osquery, BloodHound, Burp Suite, Metasploit, Nessus, OpenVAS

PROFESSIONAL EXPERIENCE

SOC Analyst — Jan 2026 – Apr 2026
ITSOLERA Pvt Ltd, Office 1786, Street 12, I-14/4 I-14, Islamabad, 46000, Pakistan

- Operated Wazuh SIEM for real-time log correlation, file integrity monitoring (FIM), and threat detection across enterprise systems.
- Configured and hardened pfSense and OPNsense firewalls, improving perimeter defense and secure VPN connectivity using WAZUH.
- Performed packet-level network analysis using Wireshark and Tcpdump to investigate anomalies and lateral movement.
- Monitored and responded to security alerts, supporting incident triage, investigation, and escalation workflows.
- Utilized tools such as Sysmon, Osquery, and Velociraptor for endpoint visibility and forensic investigation.

Cyber Security Engineer — Nov 2025 – Feb 2026
CybeCloud, Islamabad

- Conducted vulnerability assessments and penetration testing across network and web environments.
- Monitored security telemetry and investigated incidents, reconstructing attack timelines and recommending remediation.
- Tuned IDS/IPS rules and improved detection pipelines, reducing false positives and enhancing alert quality.
- Developed automation scripts to streamline repetitive security tasks and improve operational efficiency.

Campus Ambassador — Sep 2025 – Dec 2025
Hackviser Pvt Ltd, 71-75 Shelton Street, Covent Garden, London, United Kingdom(Remote)

- Delivered hands-on CTF training sessions simulating real-world cyber attacks.
- Conducted phishing awareness simulations to educate users on social engineering threats.

Cyber Security Analyst — Aug 2025 – Sep 2025
Redynox, New Delhi, India(Remote)

- Performed network scanning and vulnerability assessments using Nmap and Nessus.
- Supported penetration testing engagements and attack path analysis using Metasploit and BloodHound.
- Assisted in SOC monitoring and incident detection using SNORT-based systems.

PROJECTS & RESEARCH

ARTP & AutoPENT-Bench (Under Review at MCETS, Cyprus – First Author) 2026 — GitHub

- Designed and implemented ARTP (Autonomous Red-Team Planner), a safety-constrained offensive security system that transforms multi-source reconnaissance (Nmap, BloodHound, web crawlers) into structured, multi-stage attack plans.
- Engineered a rule-based planning engine mapped to MITRE ATT&CK, enabling automated kill-chain generation across Web, Active Directory, and Cloud environments.

- Developed a symbolic verification layer with constraint enforcement and human-in-the-loop gating to prevent unsafe or out-of-scope attack execution.
- Built AutoPENT-Bench, a reproducible benchmarking framework with standardized metrics (coverage, path efficiency, stealth, safety) for evaluating autonomous pentesting agents.
- Modeled attack execution as a state-transition system with deterministic simulation, enabling controlled evaluation of exploitation paths and agent behavior.
- Conducted large-scale experimental evaluation (N=30 runs) across containerized environments (DVWA, Juice Shop, AD simulation, cloud emulation), analyzing attack success rates and failure modes.
- Identified key automation risks including exploration drift, over-aggressive probing, and verifier dependency, contributing to safer autonomous security system design.
- Achieved high path efficiency (0.85) in Active Directory attack graphs, successfully identifying optimal privilege-escalation chains to domain dominance.

Zero-Knowledge State Migration for Post-Quantum Cryptography (BCRA elsvier (Under Review) – First Author) 2026 —

- Engineered a Zero-Knowledge proof system for securely migrating cryptographic states from classical ECDSA-based infrastructures to post-quantum cryptographic architectures.
- Designed and implemented a Groth16 zk-SNARK migration circuit, successfully compiling over 4.2 million constraints while recording proof generation and verification performance.
- Benchmarked proof generation, verification latency, CPU utilization, and memory consumption to evaluate the practicality of large-scale cryptographic state migration.
- Evaluated migration performance against current NIST Post-Quantum Cryptography standardization requirements, demonstrating compatibility with emerging quantum-safe deployment models.
- Developed a reproducible experimental framework for assessing the scalability, correctness, and computational overhead of zero-knowledge-based cryptographic transitions.

Simulating a Full-Scale Cyber Warfare in a Virtualized Datacenter 2025 — GitHub

- Designed a virtualized cyber range using Proxmox Servers, pfSense, Suricata, and Splunk SIEM.
- Implemented VLAN segmentation and simulated attack-defense scenarios.
- Built automated alerting and response pipelines for SOC operations.

ESP32-Based Offensive Security Toolkit (Flipper Zero Inspired) 2024 – 2025

- Engineered a custom ESP32-S3 based offensive security device emulating Flipper Zero capabilities for wireless attacks and hardware-level security testing.
- Developed Wi-Fi auditing modules including deauthentication attacks, handshake capture (WPA/WPA2), rogue AP deployment, and real-time network reconnaissance.
- Implemented BLE and RF-based scanning and interaction modules for detecting and analyzing nearby wireless devices and communication patterns.
- Integrated credential harvesting and packet sniffing mechanisms using low-level network interfaces and custom firmware.
- Designed modular firmware architecture in C/C++ enabling extensibility for additional attack vectors and payload execution.
- Built OLED-based user interface with real-time attack control, logging, and payload selection for field operations.
- Leveraged SD card storage for payload management, captured data logging, and offline analysis workflows.
- Simulated real-world red team scenarios including Evil Twin attacks, MITM setups, and wireless intrusion testing.
- Optimized power consumption and hardware efficiency for portable, field-deployable offensive operations.

CERTIFICATIONS

- Certified in Cybersecurity (CC) – ISC2
- Certified Cybersecurity Educator Professional (CCEP)
- Certified Threat Intelligence & Governance Analyst (CTIGA)
- Palo Alto Network Cybersecurity
- Google IT Support
- Certified Penetration Tester Specialist (CPTS) (In Progress)
- Certified Associate Penetration Tester (CAPT) (In Progress)

ACHIEVEMENTS

- FAST Intra-University Project Exhibition 2025 – 1st Position
- FAST Intra-University Project Exhibition 2026 – 1st Position
- National Idea Cup (NCCS) 2025 – 3rd Position

- DAIRA National CTF 2025 – 3rd Position
- Innovate Pakistan 2026 – Project Shortlisted
- DAMUN 2025 (UNGA Committee, DAIRA) – 1st Position (Best Delegate)
- AMUN 2025 (UNSC Committee) – 2nd Position (Outstanding Delegate)
- NUMUN 2025 (UNSC Committee, FAST Lahore) – 2nd Position (Outstanding Delegate)
- AMUN 2026 (UNSC Committee) – 1st Position (Best Delegate)
- FMUNC 2026 (UNSC Committee) – 1st Position (Best Delegate)

EDUCATION

FAST National University (NUCES)
BS Computer Science